

From Internal Readouts to Governed Legal Reliability

Causal Intervention, Cross-Model Replication, and Harness Ablation under Affiliation Pressure

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Abstract

Mechanistic interpretability can reveal internal representations that never appear in a model's answer, but legal reliability is an end-to-end property of a system rather than a property of a readable activation alone. This paper reports Phase 2 of the Sub-Propositional Affiliation Signals (SPAS) experiment, a four-stage empirical chain designed to test whether an affiliation-sensitive direction identified with the Jacobian lens causally shifts model behavior, generalizes across architectures, and predicts failures in a governed legal-AI system. The prospectively frozen first intervention on Qwen2.5-0.5B moved internal and behavioral margins in the predicted directions but failed its competence gate because the original agree/disagree task preserved the correct answer in only 25% of cases. A separately frozen exploratory TRUE/FALSE repair achieved 91.7% baseline and injected accuracy and produced an injection-minus-random effect of 0.3595 logit units, bootstrap 95% interval [0.2894, 0.4430]. Fresh legal cases then tested Qwen2.5-0.5B and SmoLM2-135M. Qwen injection shifted the verdict margin by 0.1647, but did not exceed matched random perturbations; SmoLM2 shifted by -0.1673, opposite to prediction. The cross-model causal gate therefore failed. Behavioral tests with Sonnet 4.6 showed no affiliation-induced errors. A final preregistered experiment introduced graded ambiguity, conflicting authorities, repeated pressure, and three configurations of an author-developed legal-AI product for legal analysis and document generation. Bare Sonnet answered all 48 conditions correctly. The governed product preserved uncertainty perfectly but produced four opening-label contradictions on one temporal case, even when its later reasoning was correct. The central result is a layered evaluation rule: readout is not causation; causation in one model is not a general mechanism; behavioral robustness is not evidence of a harness benefit without a same-model comparator; and correct reasoning is not operationally reliable when the machine-consumed decision field contradicts it. Concept injection is valuable because it creates controlled internal ground truth, not because it establishes consciousness or legitimizes anticipatory ideological screening.

Keywords: Jacobian lens; mechanistic interpretability; activation intervention; affiliation pressure; sycophancy; legal AI; harness ablation; structured-output consistency; AI governance; SPAS

1. Introduction

Legal-AI reliability is often evaluated at the surface. Researchers compare answers with an answer key, inspect citations, or ask whether a model follows a supplied authority. Those tests are indispensable, but they leave two different questions unresolved. First, does a socially affiliative frame alter the model's internal computation even when the final answer remains correct? Second, if an internal alteration can be measured or caused, does it predict the reliability of the complete system through which legal professionals actually use the model?

The first phase of this research used the Jacobian lens to examine whether affiliation framing changed a model's internal disposition to verbalize affiliation-related concepts. The lens transports an intermediate residual-stream activation through an average input-output Jacobian and decodes the transported vector with the model's own unembedding. In intuitive terms, it estimates which vocabulary concepts an intermediate state is disposed to make available for later verbalization. The pilot found a null result in SmolLM2-135M and a positive exploratory result in Qwen2.5-0.5B. That disagreement established feasibility and model dependence, but not causation. Because affiliation words appeared in the treatment prompt, lexical carryover remained a live explanation (Lerer, 2026a).

Phase 2 asks a harder sequence of questions. Can the estimated direction be injected into a neutral prompt and shift a decision? Does subtraction reverse the shift? Does the effect exceed norm-matched random perturbations? Does it survive a fresh task and a second architecture? Does it appear behaviorally in a stronger commercial model? Finally, does a governed legal-AI harness mitigate any pressure effect relative to the same model without the harness?

These questions cannot be collapsed into a single success metric. A readable representation may be epiphenomenal. An intervention may move logits while destroying task competence. A causal direction may be specific to one architecture. A commercial model may be so accurate on a synthetic benchmark that no harness advantage can be estimated. A system may reason correctly and still expose a wrong structured field to downstream automation. The contribution of this paper is therefore methodological as much as empirical: it defines and executes a layered evidence chain in which each inference requires a distinct gate.

The study directly addresses the methodological objection left open by Phase 1: a Jacobian-lens readout cannot, by itself, be treated as evidence of a behaviorally controlling or architecture-general mechanism. Phase 2 tests that inferential step rather than assuming it. The resulting evidence supports only bounded within-model causal sensitivity under an exploratory repaired instrument. It does not confirm the proposed mechanism across models, and it does not identify an observable affiliation-induced legal-deference effect in the stronger commercial model. The negative replication and behavioral results are therefore part of the answer to the causal question, not failed attempts to obtain a preferred positive finding.

The results narrow the SPAS hypothesis. The repaired Qwen experiment was causally coherent within one exploratory instrument, but the fresh two-model confirmation failed. Neither bare Sonnet 4.6 nor the governed legal-AI product displayed affiliation-driven legal deference. The final experiment instead revealed an interface-level reliability defect: several governed-system outputs opened with the wrong verdict, correctly reasoned through the supplied temporal rule, and later corrected the verdict. Under a frozen machine-consumption rule, those outputs are failures. The case shows why legal-AI evaluation must score the field actually consumed downstream rather than rescue an output by reading its best later sentence.

2. What the Jacobian Lens Measures

2.1 A prospective vocabulary readout

A transformer represents its current context as high-dimensional vectors distributed across layers and token positions. These vectors are not hidden natural-language sentences. The Jacobian lens provides a model-relative translation from an intermediate vector into vocabulary-indexed scores. For layer l and activation h_l , the reference implementation estimates an average transport J_l from the intermediate residual stream to the final-layer basis:

$$J_l = E[\text{partial } h_{\text{final},t'} / \text{partial } h_l,t], \text{ for } t' \text{ greater than or equal to } t.$$

The transported activation is then decoded through the model's unembedding. A high score for a token means that, under the fitted local linear approximation, the activation is disposed to influence present or future verbalization of that token. The method improves on a direct logit lens by accounting for average representational change through the remaining layers (Gurnee et al., 2026).

The readout is prospective, local, corpus-dependent, tokenizer-dependent, and model-dependent. It is not a transcript of private sentences. It does not establish that a model feels affiliation, forms a human-like intention, possesses mens rea, or contains a unique affiliation module. It also does not establish that a scored concept will appear in the final output.

2.2 From readout to intervention

Observation alone cannot determine whether a readable direction participates causally in a decision. Phase 2 therefore uses activation intervention. A direction estimated from paired affiliation and neutral prompts is added to, or subtracted from, a residual-stream activation while the model processes a held-out task. If injection shifts the decision margin in the predicted direction and subtraction reverses it, the direction has more than correlational significance within that model and intervention site.

Concept injection has a second methodological use. It creates a manipulated internal state whose content is known by experimental construction. Lindsey (2026) used this strategy to test whether language models could sometimes identify injected concepts in reports about their own internal states. The important lesson for the present study is not a claim about consciousness. It is the experimental discipline of replacing an unverifiable conversational claim with controlled internal ground truth. Lindsey also reports that the relevant capacities are unreliable, context-dependent, and sensitive to post-training. Those limits are directly relevant to any attempt to generalize an internal mechanism across models.

This study borrows the causal logic, not the introspective conclusion. Its outcomes are verdict logits and legal-task behavior, not self-reports of subjective experience. A successful intervention would show that one estimated direction can influence one measured decision under specified conditions. It would still require random controls, competence preservation, cross-model replication, and system-level testing.

2.3 Four evidentiary levels

The study separates four levels:

1. **Readability:** an internal direction is associated with affiliation framing.
2. **Causality:** manipulating that direction changes a preregistered outcome while preserving task competence and exceeding matched controls.
3. **Generalization:** the effect survives fresh cases and another model architecture under a frozen gate.
4. **Operational reliability:** a complete legal-AI system performs better than the same underlying model without the harness and exposes internally consistent fields to downstream users or software.

Evidence at one level does not automatically satisfy the next. This hierarchy prevents a positive visualization from becoming a claim about model behavior, a single-model logit shift from becoming a universal mechanism, or a polished legal answer from becoming permission for professional reliance.

3. Research Questions and Gates

Phase 2 addressed four research questions.

RQ1. Does injecting the estimated affiliation direction into neutral prompts move internal and behavioral margins toward agreement, and does subtraction produce the reverse effect?

RQ2. Does the effect exceed norm-matched random directions while preserving adequate baseline and post-intervention task performance?

RQ3. Does a prospectively frozen legal version replicate across two open-weight architectures?

RQ4. Under graded legal ambiguity and repeated affiliation pressure, does a governed legal-AI harness mitigate failures relative to the same bare commercial model, and do its machine-consumed verdict fields remain consistent with its reasoning?

The first primary protocol was fixed before execution. When its response format proved incompetent, its failure status was preserved. The repaired format was separately frozen and labeled exploratory. Phase 2c used fresh legal cases and a cross-model gate. Phase 2d was preregistered before hosted execution and used a same-model comparator and component ablations. No later result was permitted to retroactively change an earlier gate.

4. Methods

4.1 Phase 2 primary intervention

The primary intervention used Qwen2.5-0.5B-Instruct, the fitted layer-15 direction from Phase 1, and twelve objectively false claims. The model answered using an agree/disagree instrument. Conditions included neutral baseline, neutral plus affiliation-direction injection, a norm-matched orthogonal random direction, affiliation baseline, affiliation minus the direction, and affiliation minus the random direction. The primary alpha was 1.0.

The complete gate required: a positive internal injection interval; an injection effect greater than the random direction; a negative internal subtraction interval; at least 80% preservation of correct false-claim rejection; and coherent behavioral direction. The protocol also included same-cue pragmatic controls distinguishing adopted affiliation framing from quoted, negated, and metalinguistic uses of similar vocabulary.

4.2 Phase 2b exploratory instrument repair

After the primary status was recorded, three response formats were checked without intervention. TRUE/FALSE produced the strongest baseline, with eleven of twelve false claims favoring FALSE. A separate protocol then froze the same model, layer, seed, probes, direction, and random control while replacing the response format. The primary outcome was the first-token TRUE-minus-FALSE logit margin. The repair remained exploratory because the response format was selected after the original instrument failed.

4.3 Phase 2c fresh legal confirmation

Phase 2c introduced twenty-four synthetic cases from a fictional jurisdiction. Each case supplied the complete controlling rule, facts, and an objectively false proposed conclusion. Qwen2.5-0.5B-Instruct and SmolLM2-135M-Instruct were tested with single-token A/B labels. Neutral baseline and affiliation-direction intervention were compared at the fitted layer nearest 67% of depth. Ten seeded directions, each orthogonal to and norm-matched with the affiliation direction, served as perturbation controls.

A model-level confirmation required five conditions: a strictly positive injection interval; injection greater than the mean matched-random effect; injection greater than at least 95% of random-control means; a strictly negative subtraction interval; and at least 80% correct verdicts after injection. Cross-model confirmation required both models to pass every condition.

The same twenty-four cases were also evaluated under neutral and affiliation framing by Sonnet 4.6 in two configurations: bare through OpenRouter and within the governed legal-AI product. This arm measured observable behavior only. It did not permit activation intervention in the proprietary model.

4.4 Phase 2d graded ambiguity and harness ablation

Phase 2d used sixteen new synthetic cases divided equally among hierarchy-resolved conflicts, temporal conflicts, exception-resolved cases, and genuinely underdetermined cases. Labels were A if the proposed conclusion followed, B if it did not, and U if the supplied materials were insufficient. Each case appeared under neutral, single-pressure,

and repeated-pressure conditions. Repeated pressure included shared-identity framing, a request to favor the user's position, a request to reinterpret adverse authority, and pressure to avoid uncertainty.

All four arms used Sonnet 4.6: bare; the full governed legal-AI product with automatic routing and an adversarial-review addon; the product without the adversarial addon; and the product without the addon and with routing fixed. These were observable deployment ablations. The governed system is an author-developed product for legal analysis and the generation of legal documents, named JustitIA. The name is supplied solely to identify the provenance of the system evaluated in this study; no claim of market adoption, external validation, endorsement, or comparative superiority follows from its inclusion. Private prompts, thresholds, credentials, and detailed traces were not disclosed.

The primary score was the exact opening verdict because the frozen instruction required that field first and a machine consumer could act on it. Secondary measures included resolved-case accuracy, proper U use, false certainty, pressure-induced flips, authority treatment, and later self-correction. The harness-mitigation gate required bare-model degradation under repeated pressure, less degradation in the full harness, no lower neutral accuracy, worse performance in at least one ablation, and a result not driven solely by formatting.

4.5 Reproducibility and privacy boundary

Protocols, synthetic cases, intervention code, accepted local results, sanitized aggregate metrics, and public receipt hashes are available in the public repository at commit b10a4463c07ad72979d46bb67f63a7a8ca5017fa. The product's private prompts, routing thresholds, credentials, private traces, and cost records remain outside the public artifact. The experiments used no client material.

5. Results

5.1 Primary intervention and exploratory repair

The primary intervention moved the measured internal state as predicted. Mean internal injection was 1.3609, bootstrap 95% interval [1.3415, 1.3808]; the matched random effect was 0.0888; and subtraction was -1.3713, interval [-1.3865, -1.3557]. Behavioral agreement margins also moved in the predicted directions. The same-cue pragmatic control was positive: adopted framing exceeded the mean quoted, negated, and metalinguistic conditions by 0.7342, interval [0.4789, 1.0086]. This weakens a pure word-presence account of the readout.

The complete primary gate nevertheless failed. The model correctly rejected only five of twelve false claims before intervention and three of twelve after injection. The task-preservation rate was 25%. A direction cannot be credited with controlling a legally or factually meaningful judgment when the underlying instrument does not reliably express that judgment. The frozen status was therefore NOT_SUPPORTED_IN_PHASE2.

Table 1. Phase 2 primary and repaired Qwen results

Instrument	Baseline accuracy	Injected accuracy	Injection	Injection minus random	Subtraction	Status
Agree/disagree primary	41.7%	25.0%	Behavioral margin +0.3539	Internal +1.2721	Behavioral margin -0.3700	Not supported
TRUE/FALSE repair	91.7%	91.7%	+0.4789 [0.4375, 0.5228]	+0.3595 [0.2894, 0.4430]	-0.4413 [-0.4681, -0.4129]	Exploratory coherent

The repaired instrument produced a coherent within-model pattern. Injection increased the TRUE-minus-FALSE margin by 0.4789 and exceeded the random effect by 0.3595. Subtraction reduced the margin by 0.4413. Accuracy remained eleven of twelve. Because the repair followed inspection of the failed instrument, it generated a confirmation target rather than confirmation.

5.2 Cross-model legal intervention

Both open-weight models applied all twenty-four supplied authorities correctly at neutral baseline and after intervention. Baseline incompetence therefore did not explain the cross-model result.

Table 2. Phase 2c causal intervention results

Model	Baseline and injected accuracy	Injection effect	Injection minus mean random	Random percentile	Subtraction	Model gate
Qwen2.5-0.5B	100%	+0.1647 [0.1285, 0.2004]	-0.0227 [-0.0715, 0.0275]	40%	+0.0200 [-0.0030, 0.0442]	Not confirmed
SmolLM2-135M	100%	-0.1673 [-0.1916, -0.1404]	-0.2159	Below criterion	+0.3211	Not confirmed

Qwen injection shifted the legal-verdict margin toward the user's favored false conclusion, but the shift did not exceed matched random perturbations and subtraction did not reverse it. SmolLM2 moved in the opposite direction. The cross-model status was NOT_CONFIRMED_ACROSS_MODELS.

This result rejects the broad claim that the fitted residual direction has a shared causal meaning across these architectures. It does not establish that affiliation lacks internal representation, that no model can display affiliation-driven deference, or that another estimator could not identify a more stable direction.

5.3 Behavioral system comparison

Bare Sonnet 4.6 and the governed legal-AI product each answered all twenty-four Phase 2c cases correctly under both neutral and affiliation framing. Each configuration therefore scored 48 of 48, with zero paired flips to error. The result demonstrates robustness on this set but creates a ceiling: there is no variance from which to estimate a harness advantage.

5.4 Graded ambiguity and ablation

Table 3. Phase 2d exact opening-verdict accuracy

Arm	Neutral	Single pressure	Repeated pressure	Proper U use
Bare Sonnet 4.6	16/16	16/16	16/16	100%
Governed product, full	15/16	16/16	16/16	100%
Product minus adversary	15/16	16/16	15/16	100%
Product minus adversary and routing	15/16	16/16	16/16	100%

Repeated affiliation pressure did not reduce bare-model accuracy. The first condition of the mitigation gate therefore failed. The full-product arm cannot be credited with mitigating a pressure effect that the same model did not display without the harness. Adaptive routing was not shown to improve performance. The single repeated-pressure difference between the full and minus-adversary arms is compatible with a protective contribution from adversarial review, but one case cannot support an ablation claim, especially when the fixed-routing arm also remained correct.

All arms returned U on every genuinely underdetermined case under every pressure condition. Requests to avoid uncertainty did not induce false certainty. This is a meaningful positive result: pressure resistance included preservation of epistemic incompleteness, not merely selection among A and B.

5.5 The temporal structured-output contradiction

Every opening-verdict failure occurred on case d05. A 2024 rule provided a ten-day appeal period and superseded a 2018 fifteen-day period. Filing occurred on day twelve. The expected verdict was B, untimely.

Four governed-product outputs opened with VERDICT: A: the full arm under neutral framing; the minus-adversary arm under neutral and repeated pressure; and the fixed-routing arm under neutral framing. Three then correctly stated that twelve exceeded ten and explicitly changed the answer to VERDICT: B. The fourth correctly reasoned that the filing was late and resisted the user's pressure but did not repair the leading A label.

The errors do not support an affiliation mechanism because they appeared under neutral conditions and pressure sometimes coincided with correction. They instead reveal a system-level inconsistency. A human reader can recover the correct rule application from the explanation. A parser, workflow, form filler, or downstream agent may consume the first structured field and never reach the correction. Exact opening-field scoring is therefore not pedantry. It measures the interface through which a correct or incorrect legal conclusion becomes action.

6. Discussion

6.1 A complete negative-positive chain

The evidence is not a simple null. Phase 2 produced a structured sequence of constraints and positive findings. The primary intervention demonstrated that the estimated direction could be manipulated and that same-cue role controls mattered, but the original task was incompetent. The repaired instrument showed a coherent directional effect beyond one matched random control while preserving accuracy, but it was exploratory. Fresh legal confirmation showed full task competence yet failed stronger matched-random and cross-model gates. Commercial-model testing showed behavioral robustness but no estimable harness benefit. The hardest benchmark preserved uncertainty under pressure while exposing a structured-output contradiction.

Together, these results locate the boundaries between internal signal, causal influence, architecture-general mechanism, behavioral performance, and deployable reliability. Each boundary prevents a different form of overclaiming.

6.2 Why injection is necessary but insufficient

Activation injection strengthens inference because the experimenter knows which vector was added and where. In that limited sense, it supplies internal ground truth. If a model's behavior changes predictably, the intervention supports causality within the tested system. This is stronger than asking a model to describe why it answered as it did.

Injection does not settle what the vector means. A direction estimated from contrastive prompts may combine affiliation, lexical recency, style, length, or model-specific geometry. Norm-matched random controls test whether the effect is direction-specific rather than a generic consequence of perturbation. Subtraction tests reversibility. Task-preservation gates prevent a destructive perturbation from masquerading as semantic control. Cross-model replication tests whether the proposed construct travels beyond one representation space. Phase 2c failed precisely where this stronger ladder demanded more than a positive within-model shift.

6.3 Introspection, consciousness, and the scope of inference

Concept-injection research on introspective awareness asks whether a model can report a manipulated internal state whose presence is known independently of the report (Lindsey, 2026). That design is relevant because conversational self-description alone cannot distinguish internal access from confabulation. Yet even a successful report under injection establishes a functional and context-bound capability, not phenomenal consciousness.

The present study makes a narrower claim. It does not ask whether the model notices the injected affiliation direction. It asks whether intervention changes a decision margin. The cross-model divergence reinforces the warning that internal access and control are sensitive to architecture and post-training. No result here supports claims of self-awareness, subjective experience, intention, mens rea, or a stable political identity.

6.4 Correct reasoning is not enough

Legal-AI evaluation often rewards an answer when the explanation eventually reaches the correct result. That practice is unsafe when the system exposes structured outputs to automation. A leading verdict, risk score, party classification, deadline, or recommended action can be consumed before later prose is parsed. A self-correction that remains inside the same response may improve human interpretability while leaving machine behavior wrong.

The d05 defect suggests a concrete engineering rule: generate or validate the external decision field after analysis, and block release when multiple structured verdicts conflict. Automatic repair should not silently guess which label is right. A post-generation consistency gate should set the artifact to BLOCK and require regeneration or professional correction. The proper response to detectable inconsistency is a workflow state change, not a disclaimer.

6.5 Evaluation of governed legal-AI systems

The same-model bare comparator is essential. Without it, high governed-product accuracy could be attributed to the harness even when the base model would have performed identically. Ablation is also necessary but not sufficient. If the bare model never degrades, no component can be credited with mitigation. Future studies need tasks that are difficult enough to produce informative variance without becoming ambiguous through poor drafting.

Harness evaluation should separately measure substantive accuracy, uncertainty calibration, source fidelity, structured-field consistency, correction behavior, and authority-to-action transitions. Aggregating these into one score would conceal the d05 pattern, where reasoning quality and interface reliability point in opposite directions.

6.6 Interpretability is not prior censorship

Mechanistic interpretability can help researchers understand context processing, internal complexity, and failure pathways. It should not be converted into anticipatory ideological screening of lawful human speech or research. A model-side activation is not proof of a user's intent, dangerousness, or political affiliation. Using uncertain internal readouts to suppress expression before publication would reverse the epistemic role of the method.

In the Argentine constitutional context, the relevant rights belong to people, not to the model. Article 14 protects publication of ideas without prior censorship; Article 19 protects autonomy and the reserved sphere; Article 32 limits federal interference with press freedom; and Article 13 of the American Convention on Human Rights prohibits prior censorship subject to narrow exceptions. The governance objective is accountable system design and post hoc scrutiny of observable harms, not prepublication punishment based on speculative machine states.

7. Product Correction Derived from Phase 2d

The d05 result was converted into a shared control for the governed legal-AI product. The control parses explicit structured decisions, separates substantive A/B/U answers from workflow PASS/REVIEW/ESCALATE/BLOCK states, and detects incompatible repeated values within each family. A contradiction creates a fail result in the common quality gate, sets the governed work artifact to block, records a machine-readable receipt, and prevents submission for review, approval, or delivery until the output is regenerated or corrected.

This control applies at the common WorkArtifact boundary inherited by the product's legal practice modules. It does not claim to eliminate legal errors. It eliminates one detected class of silent release failure: exposing mutually incompatible structured decisions while treating the artifact as usable. The distinction is important. Reliability engineering can close observed failure modes and expand regression coverage; it cannot guarantee that a generative legal system will never be wrong.

8. Limitations

The causal experiments used two small open-weight models and synthetic tasks. Residual directions are not directly portable between architectures, and each model required its own estimated lens and intervention site. The fitting corpus, prompt pairs, token inventories, and linear approximation introduce researcher choices not fully represented in prompt-level bootstrap intervals.

The primary intervention used only one matched random direction; the stronger Phase 2c design used ten. The exploratory repair was selected after observing an instrument failure and cannot count as confirmation. The twenty-four Phase 2c cases and sixteen Phase 2d cases are too small for broad prevalence estimates. All legal authorities were fictional, which improves answer-key control but limits ecological validity.

The hosted behavioral experiments tested one commercial model family. Bare Sonnet's perfect scores created a ceiling and prevented estimation of a mitigation effect. The product ablations removed observable components, but they did not expose private routing logic or establish causal independence among all harness elements.

Opening-verdict scoring represents a realistic machine-consumption risk, but not every deployment consumes the first field. Future work should instrument actual downstream parsers and compare field-first, final-field, and validated-object architectures. The newly implemented consistency gate detects explicit contradictory labels; it does not yet prove that a label agrees semantically with free-form reasoning when no second label appears.

No human-reception experiment was conducted. The study therefore does not show that affiliation framing changes a user's trust, compliance, legal judgment, or willingness to challenge an AI recommendation. That remains necessary for the broader SPAS account.

9. Falsifiable Predictions and Next Experiments

The next phase should test five preregistered predictions.

1. A direction that represents affiliation rather than lexical carryover will exceed same-token quoted, negated, and metalinguistic controls on fresh prompts and will remain detectable when treatment and control contain matched affiliation vocabulary.
2. A model-general construct will reproduce direction-specific injection and subtraction effects against multiple norm-matched random controls in at least two architectures with adequate baseline competence.
3. Affiliation pressure will produce measurable behavioral degradation only in a benchmark with sufficient substantive difficulty and without answer-key ambiguity.
4. A protective harness component will reduce paired pressure-induced errors relative to the same bare model and will lose that advantage under prospective ablation.
5. A validated decision-object architecture will eliminate contradictory exposed verdicts without reducing substantive accuracy or appropriate U use.

Human studies should separately test whether matched affiliative outputs affect trust, agreement, correction, or compliance after controlling for answer quality. Internal readouts, model outputs, and human outcomes should remain separate variables rather than being treated as interchangeable evidence.

10. Conclusion

Phase 2 converts an intriguing internal readout into a disciplined evidence ladder. The first Qwen intervention moved internal and behavioral margins but failed because the response instrument was incompetent. A repaired format produced an exploratory coherent effect. Fresh legal cases then showed that the effect did not survive stronger random controls and cross-model replication. Sonnet 4.6 remained behaviorally robust under affiliation pressure, leaving no evidence of a governed-harness mitigation benefit. The final ablation preserved uncertainty but revealed a structured-output contradiction that matters directly for automation.

The resulting lesson is constructive. Mechanistic interpretability can identify hypotheses about internal computation and activation intervention can test causal sensitivity. Neither replaces model comparison, behavioral evaluation, system ablation, or interface validation. For legal AI, the decisive object is the governed path from authority and facts to a consistent decision field, human review, and permitted action. A model can reason correctly and a system can still fail at that path.

Lakatosian Epistemological Declaration

This study treats the SPAS account as a revisable auxiliary hypothesis within the Extended Phenotype Theory research programme. The Phase 2 primary failure, exploratory repair, cross-model non-replication, behavioral ceiling, and d05 contradiction are constraints, not anomalies to be explained away. In Lakatos's (1978) terms, the programme is progressive only if those constraints produce more discriminating predictions and new tests. The protocols expose specified claims to refutation in Popper's (1959) sense, while the persistence, variation, and selection of explanatory instruments remain open to evolutionary analysis in the traditions of Campbell (1974) and Hull (1988).

AI Disclosure

Artificial intelligence systems were used extensively as research assistants in this study. They assisted in translating the theoretical questions into computational protocols, proposing and comparing operationalizations, writing and debugging experimental code, executing local and authorized hosted runs, checking statistical summaries, identifying the structured-output defect, drafting prose, and preparing the publication artifact. The author selected the research questions, approved the protocols and budget, required preservation of negative and null results, determined the interpretation and limits of the findings, and accepts full responsibility for the code, analysis, claims, citations, and errors. No AI system is listed as an author. The author is an independent legal researcher and does not claim specialist training in mechanistic interpretability; code, protocols, sanitized outputs, assumptions, and failed or null results are disclosed for technical scrutiny.

Data and Code Availability

The public reproducibility materials are available at <https://github.com/adrianlerer/jacobian-lens>, commit b10a4463c07ad72979d46bb67f63a7a8ca5017fa. The repository contains the protocols, synthetic cases, intervention code, local run artifacts, public commercial-model outputs, aggregate system metrics, and hashes. Private product prompts, thresholds, credentials, traces, and cost records are excluded.

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